SMIT AI & DATA SCIENCE

Customer Churn Prediction & Retention Analytics

End-to-end machine learning system and interactive dashboard

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Live app: Streamlit Community Cloud

TOTAL CUSTOMERS

7,043

IBM Telco dataset, 21 columns

OVERALL CHURN RATE

26.54%

1,869 customers churned

CHURNED MONTHLY REVENUE

$139,130.85

Revenue already lost to churn

MONTH-TO-MONTH SHARE

86.9%

$120,847 of the churned monthly revenue

Business Context and Data Pipeline

1. Business problem

Revenue leakage: $139,130.85 of monthly revenue has churned.

Reactive retention: action happens after customers cancel.

Untargeted outreach: offers go out without a risk ranking.

Unclear drivers: which contract, billing or service factors relate to churn.

2. Cleaning and integrity

Target: Churn_Binary, 0 = retained (73.46%), 1 = churned (26.54%).

TotalCharges: text column; 11 blank values (tenure 0) set to 0, no rows dropped.

Leakage control: customerID removed; preprocessing fitted on training data only.

Selection: by ROC-AUC and recall, not accuracy.

3. Eight engineered features

Tenure_Group: ≤12, 13–24, 25–48, 49–72 months.

Monthly_Charge_Group: four charge bands.

Customer_Segment: Family, Solo, Senior Family, Senior Solo.

Contract_Level: 0, 1, 2 by contract length.

Unprotected_Services_Count: missing add-ons (internet customers).

has_internet: 1 if internet service.

Payment_Risk: electronic check High, mailed check Medium, automatic Low.

Customer_Value: tenure × monthly charges.

EDA: Where Churn Concentrates

CONTRACT TYPE

42.71% vs 2.83%

Month-to-month vs two-year churn, about 15 times higher. Month-to-month churners carry $120,847 (86.9%) of churned monthly revenue.

PAYMENT METHOD

34.7% vs 16.0%

Manual payers vs auto-pay. The gap is 13.0 points on month-to-month and 0.1 points on two-year contracts, so contract explains most of it.

TENURE

47.44% vs 9.51%

New customers (12 months or less) vs loyal customers (49–72 months). Churn falls steadily as tenure grows.

PROTECTION SERVICES

41.6% vs 15.2%

Internet customers without vs with Tech Support. Online Security shows the same pattern: 41.8% vs 14.6%.

Model Evaluation on the Held-Out Test Set

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest (selected)	**0.804**	**0.665**	**0.527**	**0.588**	**0.844**
Logistic Regression	0.797	0.646	0.521	0.577	0.843
Decision Tree	0.797	0.668	0.468	0.550	0.838
K-Nearest Neighbors	0.777	0.582	0.570	0.576	0.821

Why Random Forest was selected

Highest ROC-AUC (0.844), although the four models are close (Logistic Regression 0.843).

Accuracy is not the basis: only 26.54% of customers churn.

Risk tiers separate well: actual churn is 8.2% in the Low tier and 79.1% in the Critical tier.

Confusion matrix (threshold 0.50)

TN 936 — Retained, correct	**FP** 99 — Wrongly flagged
FN 177 — Missed churners	**TP** 197 — Churners caught

1,409 test customers: 374 churned, 1,035 retained.

Threshold Tuning: Recall vs Outreach Capacity

Matching the cutoff to capacity

0.30, high recall: flags 543 customers and catches 291 of 374 churners (recall 77.81%).

0.50, default: flags 296; precision 66.55%, recall 52.67%.

0.60, high precision: flags 172 with precision 74.42%, for small retention teams.

Best F1: 0.30 (63.47%) among the cutoffs shown.

Threshold	Flagged	Precision	Recall	F1	TP	FP	FN
0.20	688	46.51%	85.56%	60.26%	320	368	54
0.30	543	53.59%	77.81%	63.47%	291	252	83
0.40	419	58.23%	65.24%	61.54%	244	175	130
0.50 (default)	**296**	**66.55%**	**52.67%**	**58.81%**	**197**	**99**	**177**
0.60	172	74.42%	34.22%	46.89%	128	44	246
0.70	96	79.17%	20.32%	32.34%	76	20	298

Verified on the 1,409-customer test set. Lower cutoffs catch more churners but flag more customers who would have stayed.

Streamlit Application: 8 Pages

1

Home

Business problem, objectives and dataset summary.

2

Customer Explorer

Filter 7,043 customers and download the filtered data.

3

Churn Dashboard

Eight KPI cards and revenue at risk by contract.

4

Churn Analysis

Ten interactive charts with segment and tenure filters.

5

Model Comparison

Metrics, confusion matrix, ROC, precision-recall, importances.

6

Threshold Analysis

Slider recomputes precision, recall, F1, FP and FN.

7

Churn Prediction

Single-customer probability, risk level and actions, with CSV.

8

Retention Dashboard

1,477 High and Critical customers, $76,429.55 expected monthly revenue at risk, CSV list.

Retention Recommendations (1 of 2)

		FINDING	EVIDENCE	RECOMMENDATION	EXPECTED RESULT
1	**Annual contract offer**	Month-to-month churn is 42.71%, about 15 times the two-year rate of 2.83%.	Churn Dashboard, revenue at risk by contract.	Offer discounted annual contracts, starting with High and Critical risk month-to-month customers.	These customers carry $120,847 of $139,131 churned monthly revenue. Retaining 20% is worth about $24,169 a month.
2	**Auto-pay switch**	Auto-pay churn is 16.0% vs 34.7% for manual payers, but the gap is 13.0 points on month-to-month and 0.1 on two-year contracts.	Churn Analysis, auto-pay chart; notebook Section 4.1.	Offer a one-time bill credit for switching to auto-pay, run separately from the contract offer.	A smaller gain than the 18.7-point raw gap suggests. Confirm with a pilot.
3	**Senior Solo outreach**	Senior Solo has the highest churn rate (49.2%) and the highest share flagged High or Critical (50.45%).	Churn Analysis, segment chart; Retention Dashboard, segment table.	Give High and Critical Senior Solo customers a named contact or proactive check-in call.	Effort goes to the most concentrated risk. Compare churn against a control group.

Recommendations 4–5 and Limitations

		FINDING	EVIDENCE	RECOMMENDATION	EXPECTED RESULT
4	**Risk-tier outreach order**	Held-out churn is 8.2% in the Low tier and 79.1% in the Critical tier. High and Critical: predicted 65.0%, observed 66.55%.	Retention Dashboard, risk tiers; notebook Section 6.	Contact Critical customers first, then High. Use the Threshold Analysis page to match capacity.	Effort goes to the customers most likely to leave.
5	**Monthly campaign with control**	About 21% of customers are High or Critical. At 0.50 the model flags 296 of 1,409 test customers.	Retention Dashboard, CSV list; Threshold Analysis.	Export the list monthly, contact a sample, and hold back a control group.	Shows which action works and how much revenue it protects.

Limitations

Association, not causation; no satisfaction, complaints, NPS, region or referral data in this dataset.

Retention Dashboard scores all 7,043 customers (in-sample): held-out result is 296 High and Critical customers, about $15,030 expected monthly revenue at risk.

Summary, Repository and Live Demo

What the project delivers

Discrimination: Random Forest reaches ROC-AUC 0.844 on the held-out test set.

Revenue exposure: $139,130.85 churned monthly revenue; $76,429.55 expected at risk across all customers, about $15,030 on the test set.

Threshold control: slider from 0.20 (recall 85.56%) to 0.70 (precision 79.17%).

Deployed app: 8 pages with live scoring and CSV downloads.

Rigor: leakage checked, limitations documented, A/B testing recommended.

Deliverables

Live app: fp-churn-tsqxrn6e86avqpcszde6rc.streamlit.app

GitHub: github.com/zaidali21082007-spec/fp-churn

Business Insight Report: methodology, findings and 5 recommendations.

Next: live demo of the threshold slider and single-customer scoring.